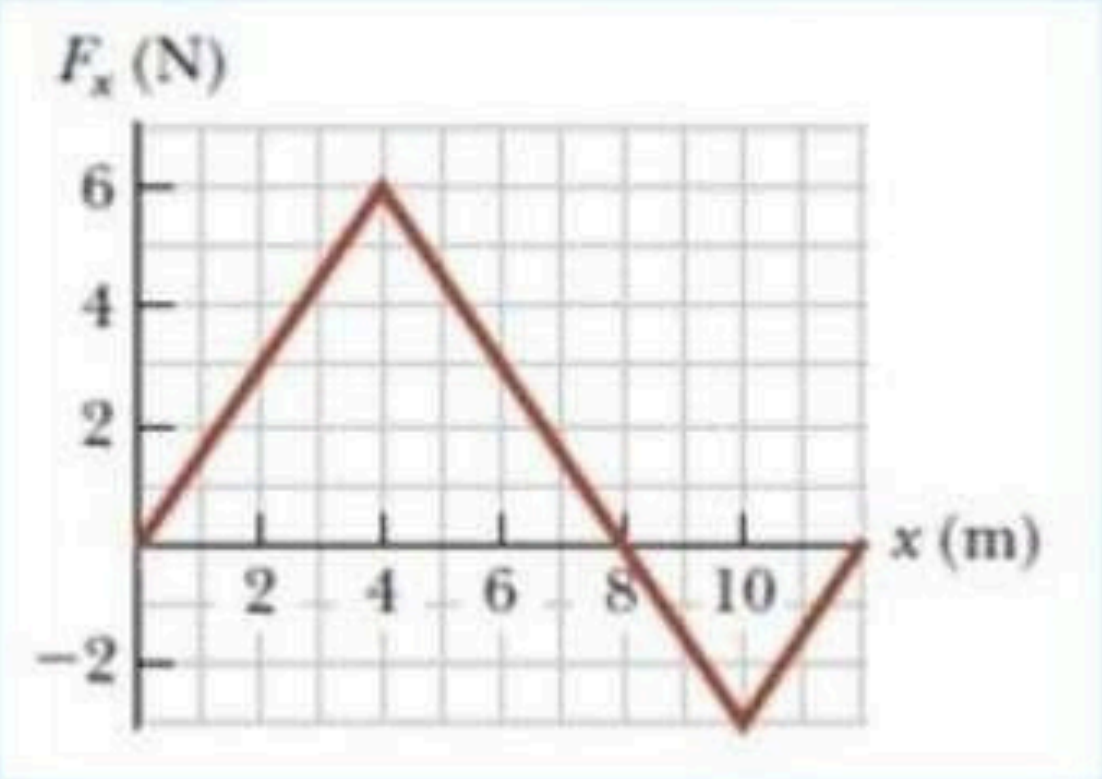
The force acting on a particle varies as shown in Figure below. Find the work done by the force on the particle as it moves from x = 0 to x = 8 m.



Select one:

- a. **0**
- b. **24 J**
- c. **-24J**
- d. **48J**
- e. **21J**

The only force acting on a 1.8 kg body as it moves along the x axis is given by $F_x = 3x$ N, where x is in m. If the velocity of the body at x = 0 is + 8 m/s, at what value of x will the body have a velocity of + 4 m/s?

اخترأحد الخيارات

a. 28.8 m/s

b. 5.37 m

c. 14.4m/s

d. 2.88m/s

e. 3.79m/s

A small mass of 10g placed 10cm from the center of a rotating, horizontal turntable starts slipping when its speed is 20cm/s. What is the coefficient of static friction between the small mass and the turntable?

Select one:

- a. 0.04
- b. 0.25
- c. 4.00
- d. 0
- e. 0.4

In a given displacement of a particle, its kinetic energy increases by 25 J while its potential energy decreases by 10 J. Determine the work of the nonconservative forces acting on the particle during this displacement.

اخترأحد الخيارات

a. −35 J

b. −15 J

c. +35 J

d. +55 J

e. +15 J

A 0.4-kg particle has a speed of 2m/s at point A and kinetic energy of 8J at point B. What is the net work done on the particle by external ?forces as it moves from A to B

اختر أحد الخيارات

a. 6 J

b. 0

c. 8.8 J

d. 12.8 J

e. 7.2 J

A block of mass 40 kg initially at rest is pushed 5.00m along a rough, horizontal surface with a constant applied horizontal force of 100N. The coefficient of kinetic friction between the block and the surface is 0.20. Find the increase in the internal energy of the system ($\Delta E_{int.}$).

اخترأحد الخيارات

a. 4J

b. -392J

c. 588J

d. 392J

e. 108J

A tire 0.500 m in radius rotates at a constant rate of 100 rev./ min. Find the linear velocity of the tire.

Select one:

- a. 5.23 m/s
- b. 3.14 m/s
- c. 2.62 m/s
- d. 314 m/s
- e. 52.33 m/s

A 4.00-kg particle moves along the x axis. Its position varies with time according to the equation of motion $X(t) = t+2t^3$ where x is in meters and t is in seconds. The magnitude of the resultant force that acts on the particle at time t=2s is:

Select one:

- a. Cannot be determined from this equation
- b. 100 N
- c. 96 N
- d. 24 N
- e. 25 N

A spring with spring constant k = 800 N/m is compressed 12 cm from its equilibrium position. A spring with spring constant k = 400 N/m has the same elastic potential energy as the first spring when its extension is:

اخترأحد الخيارات

a. 8.0cm

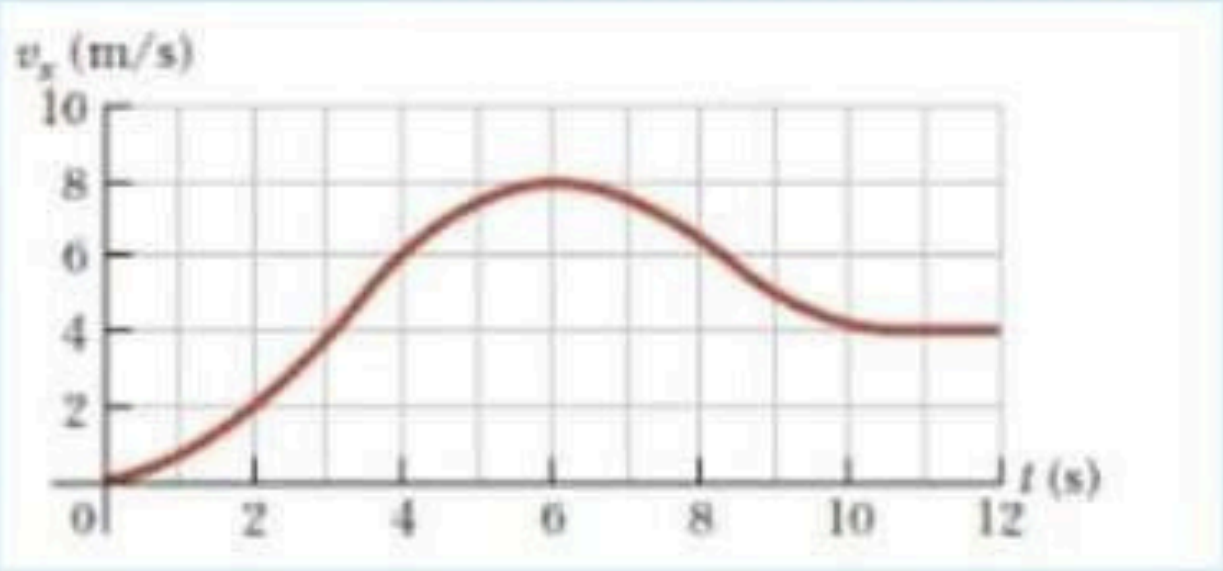
b. 5.0cm

c. 18cm

d. 17 cm

e. 6.0cm

The figure below shows a graph of v_x versus t for the motion of a person as he starts from rest and moves along the road in a straight line. At what time value, the person changes his direction?



Select one:

a. **10 s**

b. **Cannot be determined from this graph**

c. **6 s**

d. **4 s**

e. **zero**

A 2.0-kg object passes through the origin at time t=0, such that its x component of its velocity is 5.0m/s and its y component of its velocity is -3.0m/s. What is the kinetic energy of the object at this time?

Select one:

- a. 34J
- b. 9J
- c. 340J
- d. 16J
- e. 25J

An airplane travels 80 m/s as it makes a horizontal circular turn which has a 0.80 km radius. What is the magnitude of the resultant force on the 75 kg pilot of this airplane?

Select one:

a. **6 kN**

b. **600 kN**

c. **5.88 kN**

d. **60 kN**

e. **0.60 kN**

A block is given an initial velocity of 5 m/s up a frictionless incline of angle $\theta = 30^{\circ}$. How far up the incline does the block slide before coming to rest?

Select one:

- a. 2.55 m
- b. 0.51 m
- c. 1.27 m
- d. 1.00 m
- e. 0.25 m

A rock is dropped from a bridge. What happens to the magnitude of the acceleration and the speed of the rock as it falls? [Neglect friction.]

اخترأحد الخيارات

a. Acceleration increases and speed decreases.

b. Both acceleration and speed remain the same.

c. Both acceleration and speed decrease.

d. Acceleration remains the same and speed increases.

e. Both acceleration and speed increase.

A long jumper leaves the ground at an angle of 30 degree above the horizontal and at an unknown speed (v). If the maximum height reached is 1.05m. What is the value of v?

Select one:

a. 9.07 m/s

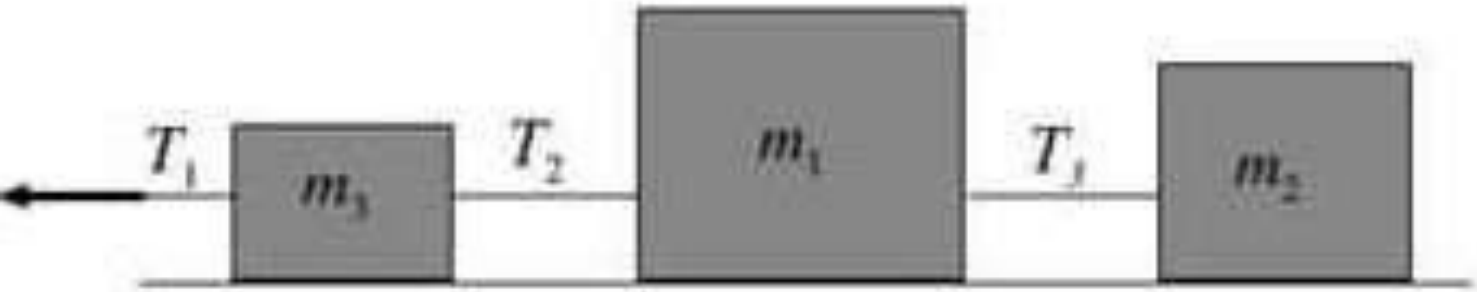
b. 82.32 m/s

c. 0.722 m/s

d. 13.31 m/s

e. 15.93 m/s

Three masses are connected by massless strings. If $m_1 > m_2 > m_3$, which of the following gives the appropriate ordering of the magnitudes of the tensions in the ?strings



اختر أحد الخيارات

a. $T_2 > T_1 > T_3$

b. $T_1 > T_2 > T_3$

c. $T_1 = T_2 = T_3$

d. $T_1 > T_3 > T_2$

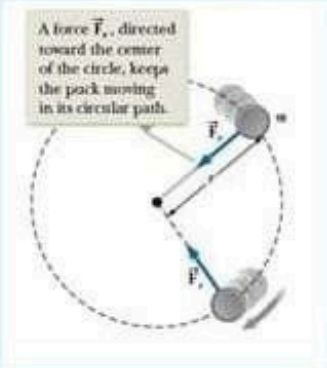
e. $T_3 > T_2 > T_1$

A ball is thrown directly downward with an initial speed of 8.00 m/s from a height of 30.0 m. After what time interval does it strike the ground?

Select one:

- a. 1.8 s
- b. 3.4s
- c. 18 s
- d. 2.0 s
- e. 34s

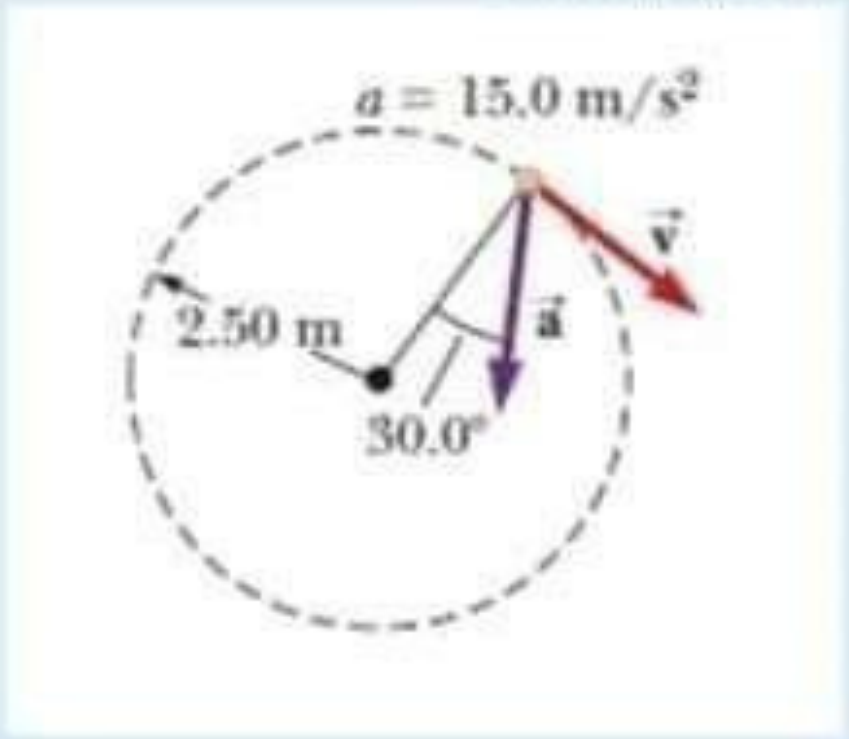
A puck of mass 0.500 kg is attached to the end of a cord 1.50 m long. The puck moves in a horizontal circle as shown below. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the puck can move before the cord breaks?



Select one:

- a. 12.20 m/s
- b. 150 m/s
- c. 14.70 m/s
- d. 0.15 m/s
- e. 3.83m/s

The figure below represents the total acceleration of a particle moving clockwise in a circle of **radius 2.5 m** at a certain instant of time. For that instant, find its tangential .acceleration



اختراحد الخيارات

a. 9.8 m/s^2

b. 3.8 m/s^2

c. 1.3 m/s^2

d. 5.7 m/s^2

e. 7.5 m/s^2

Taking A= (6.00i-8.00j) units, B= (-8.00i+3.00j) units, and C= (26.0i+19.0j) units. Determine the value of a, such that: aA+ 7B+C=0.

Select one:

- a. 5.0
- b. -5.0
- c. 13.7
- d. 5.3
- e. -13.7

An 820 N Marine in basic training climbs a 12 m vertical rope at a constant speed in 8 seconds. What is his power output?

Select one:

- a. 102.5kW
- b. 12.05kW
- c. 1.23 kW
- d. 0
- e. 6.56kW

An elevator together with its passengers weighs 4900 N. At a certain instant, the tension in its supporting cable is 5900 N. Determine the magnitude of its .instantaneous acceleration

(Use g = 9.8 m/s^2)

اختر أحد الخيارات

.a. 1.8 m/s^2

.b. 2.0 m/s^2

.c. 4.0 m/s^2

.d. 2.8 m/s^2

.e. 0 m/s^2

A constant force F = 72 N is applied to a 6 kg block, which contacts a 12 kg block. They accelerate together on a horizontal frictionless surface. What is the magnitude of the force on the 6 kg block due to the contact with the ?12 kg block

اختر أحد الخيارات

a. 24 N

b. 29 N

c. 72 N

d. 48 N

e. 4 N

A particle moves according to the equation $X(t)=10t^2$, where x is in meters and t is in seconds. What is the acceleration at t= 2s?

اختر أحد الخيارات

a. 38 m/s^2

b. 10 m/s^2

c. 30 m/s^2

d. 20 m/s^2

e. 40 m/s^2

Vector A has a magnitude of 5.00 units, and vector B has a magnitude of 9.00 units. The two vectors make an angle of 50° with each other. Find A . B

اختر أحد الخيارات

a. 30.4

b. 28.9

c. 56.4

d. 53.9

e. 60

Using the definition of the scalar product, find the angle between the next vectors.
A=3i - 2j and B= 4i - 4j

اختر أحد الخيارات

a. 45^{o}

b. 90^{o}

c. 11.3^{o}

d. 1.9^{o}

e. 0^{o}

A 6 kg block initially at rest is pulled to the right along a frictionless, horizontal surface by a constant horizontal force of magnitude 12 N. Find the blocks speed after it has moved through a horizontal distance of 3 m.

اختر أحد الخيارات

a. 6.20 m/s

b. 0 m/s

c. 8.60m/s

d. 3.50 m/s

e. 4.80 m/s

A block is given an initial velocity of 5 m/s up a frictionless incline of angle $\Theta = 30^{o}$. How far up the incline does the block slide before coming to rest? (use $g = 9.8\ m/s^{2}$)

اخترأحد الخيارات

a. 48.2 m

b. 4.5m

c. 2.55 m

d. 4.0 m

e. 3.73m

Vector A = -6i + 4j. Find the magnitude of vector B = -2i + bj, if ?B is perpendicular to A
i and j are the unit vectors in the x] [and y-direction, respectively

اختر أحد الخيارات

a. 3.0

b. -1

c. -6

d. 3.6

e. 6.0

Vector A has a magnitude of 29 units and points in the positive y direction. When vector B is added to A, the resultant vector A+B points in the negative y direction with a magnitude of 14 units. What ?is the magnitude of B

اختر أحد الخيارات

a. 34units

b. 43 units

c. 7units

d. 6units

e. 40units

An object of mass 5 kg has a kinetic energy of 100 J, then it's momentum is equal to:

اخترأحد الخيارات

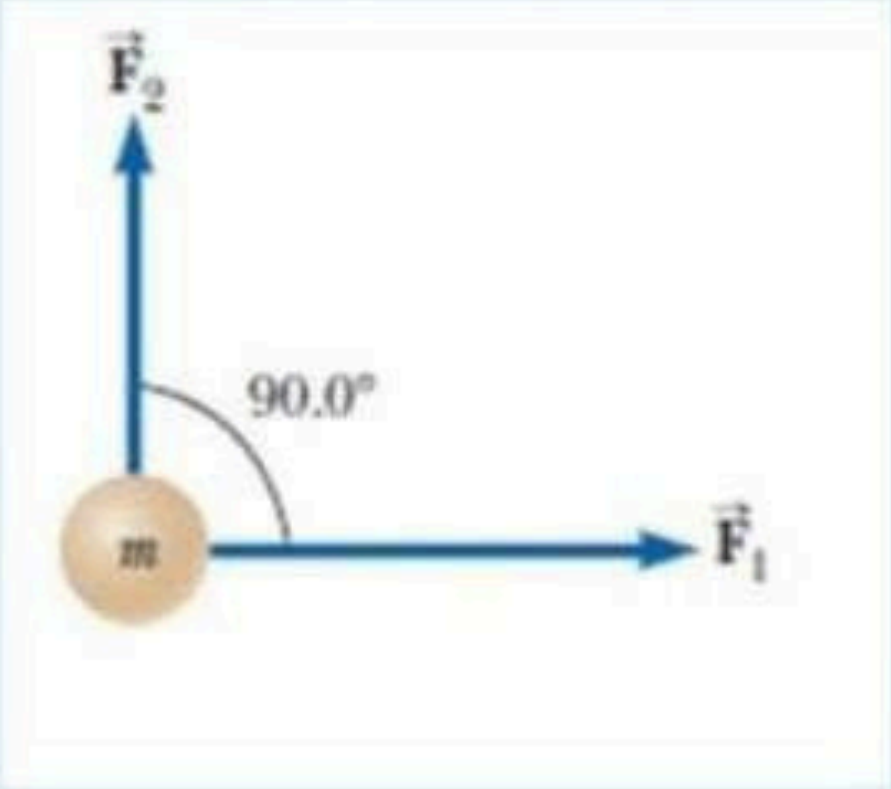
a. 40.0 kg.m/s

b. 31.6 kg.m/s

c. 50.0 kg.m/s

d. 44.7 kg.m/s

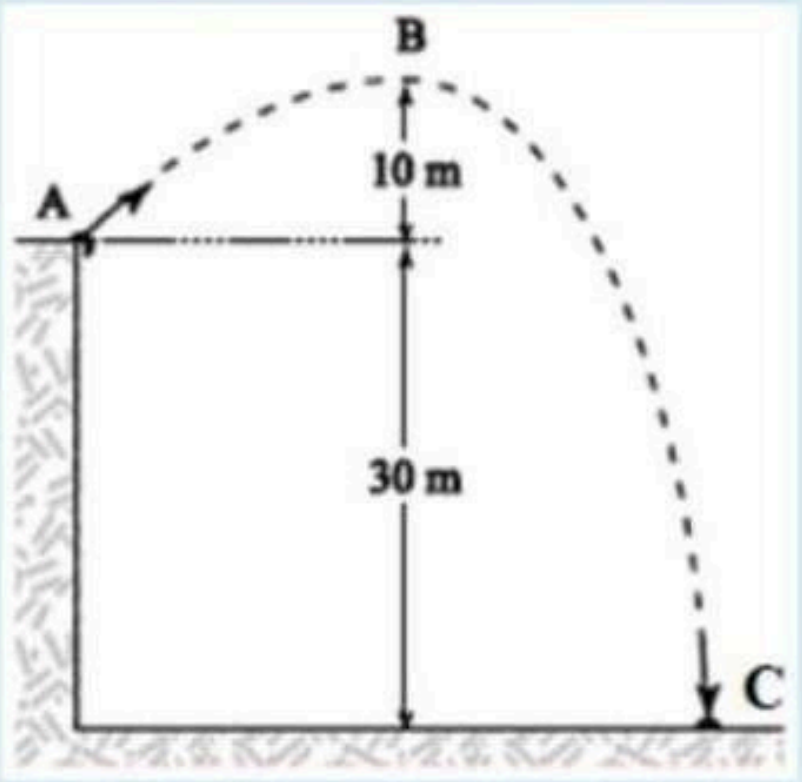
Two forces F_1 and F_2 act on a **5 kg** object. Taking F_1 = **30 N** and F_2 = **40 N**, find the acceleration of the object for the configuration of forces shown in the figure below



Select one:

- a. 26 m/s^2.
- b. 10 m/s^2.
- c. 5 m/s^2.
- d. 20 m/s^2.

A **0.40 kg** ball is thrown at an unknown angle above the horizontal from the top of tall building (point **A as** shown in Figure below) . The ball attains a maximum height at **B** before striking the ground at point **C**. What is the value of the kinetic ?energy of the ball at **C**



اخترأحد الخيارات

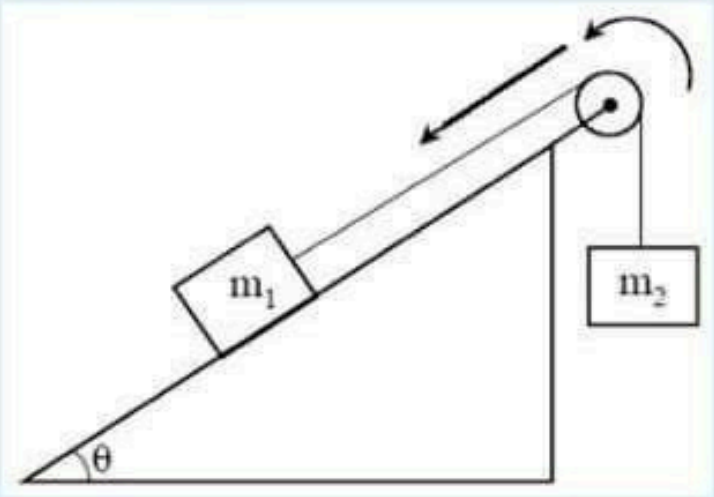
a. 196.0 J

b. 117.6 J

c. 156.8 J

d. 215.6 J

Two blocks of masss $m_1 = 20$ kg and $m_2 = 4$ kg are attached to a light rope passing over a pulley as shown in the figure. Block m_1 lies on a frictionless incline of angle $\theta = 36.9°$. The system is released from rest and moves as shown. Find **the acceleration (in m/s^2) of the two objects**.



Select one:

- a. 3.27
- b. 5.88
- c. 2.26
- d. 2.74

A man on a strange planet finds that he can jump a maximum horizontal distance of 15 m if his initial speed is 3 m/s. What is the free-falling acceleration on the planet?

(use $g=9.8\ m/s^2$)

اختر أحد الخيارات

a. $0.7\ m/s^2$

b. $1\ m/s^2$

c. $0.6\ m/s^2$

d. $1.6\ m/s^2$

e. $0.5\ m/s^2$

A car covers 40.0 m in 8.50 s while smoothly slowing down to a final speed of 2.80 m/s. What is its ?acceleration

اختر أحد الخيارات

a. $-0.38\ m/s^2$

b. $-0.45\ m/s^2$

c. $40\ m/s^2$

d. $-45\ m/s^2$

e. $-40\ m/s^2$

A car leaves a stop sign and exhibits a constant acceleration of $0.3\ m/s^2$ parallel to the roadway. The car passes over a rise in the roadway such that the top of the rise is shaped like an arc of a circle of radius 500 m. At the moment, the car is at the top of the rise, its velocity vector is horizontal and has a magnitude of 6 m/s. What is the magnitude of the radial acceleration for the car at this instant?

اخترأحد الخيارات

a. $5.7\ m/s^2$

b. $1.3\ m/s^2$

c. $0.072\ m/s^2$

d. $0.038\ m/s^2$

e. $0.085\ m/s^2$

Vector A = -6i + 4j. Find the magnitude of vector B = -2i + bj, if ?B is perpendicular to A
i and j are the unit vectors in the x] [and y-direction, respectively

اختر أحد الخيارات

a. 3.0

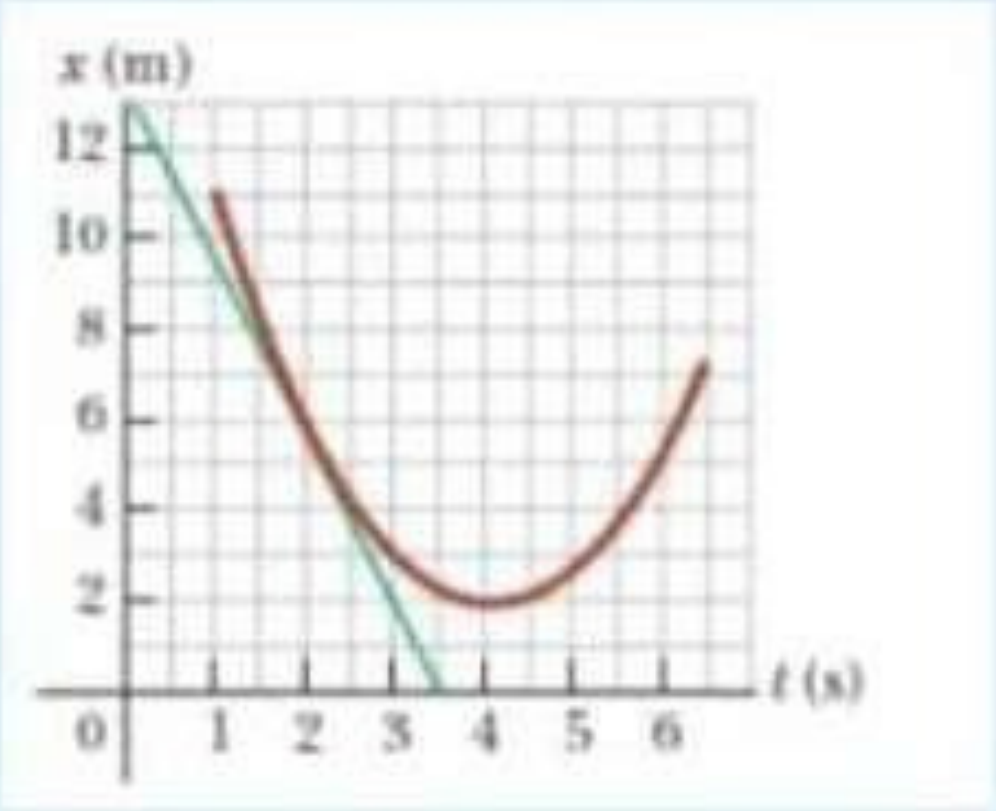
b. -1

c. -6

d. 3.6

e. 6.0

A position-time graph for a particle moving along the x-axis is shown in the next figure. What is the ?velocity at t= 2 s



اخترأحد الخيارات

a. -3.71m/s

b. Cannot be determined from the graph

c. -3.8 m/s

d. 3.71 m/s

e. 3.8 m/s

The following reaction is observed:

$$2A + bB \rightarrow cC$$

As the reaction progressed, the concentrations changed by these rates,

$rate_A = 0.050$ mol/L·s

$rate_B = 0.150$ mol/L·s

$rate_C = 0.075$ mol/L·s

What are the values for the coefficients b, and c respectively?

Select one:

- 6,1
- 3,6
- 6,2
- 6,3
- 3,3

A spring with spring constant **k = 1000 N/m** is compressed **20 cm** from its equilibrium position. A spring with spring constant **k = 500 N/m** has the same elastic potential energy as the first spring when its extension is

اخترأحد الخيارات

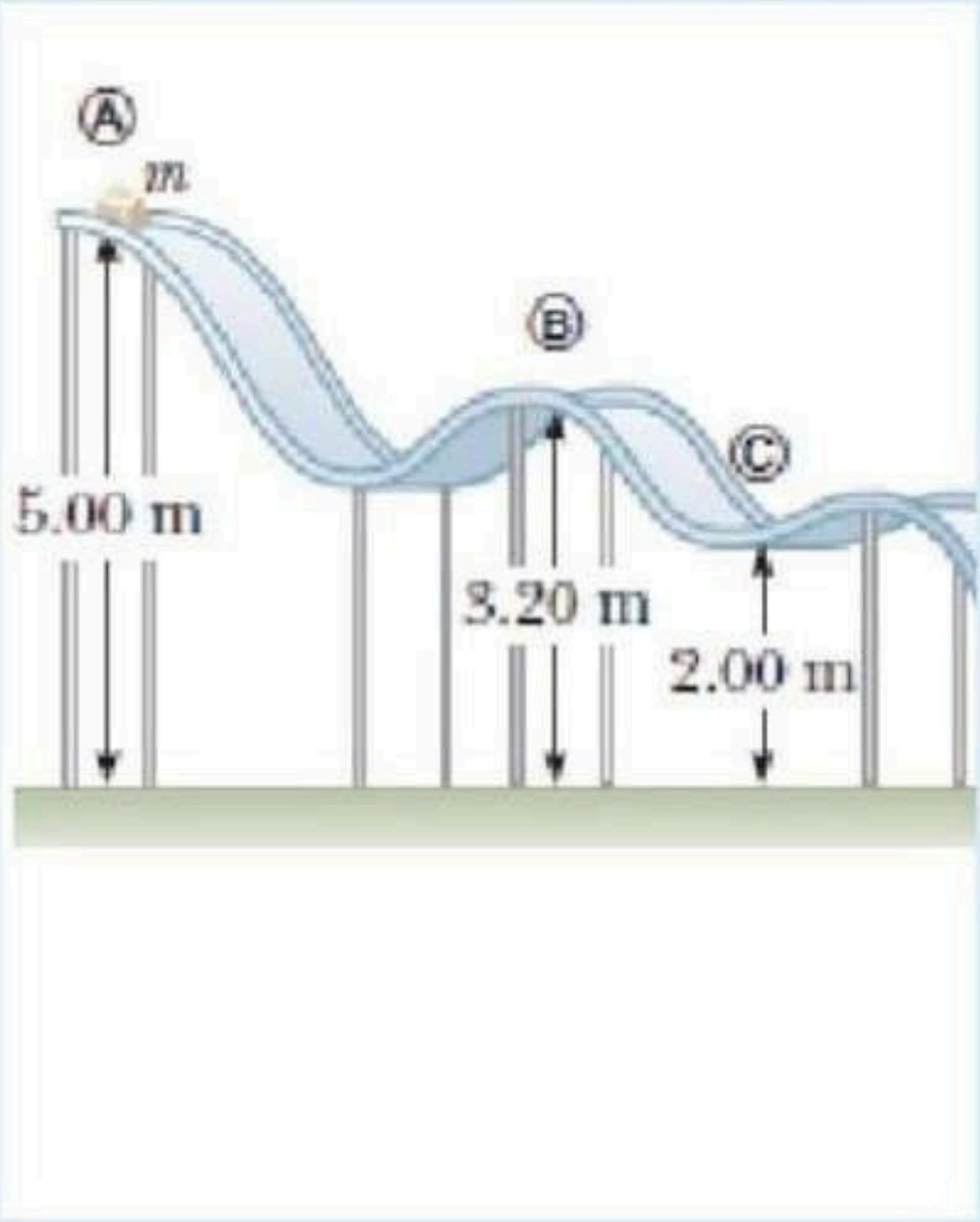
.a. 0.14 m

.b. 0.28 m

.c. 0.23 m

.d. 0.17 m

A block of mass m=5.0 kg is released from point A and slides on a frictionless track as shown on the figure below. Determine the net work done on the block by the gravitational force as it moves from A to C.



Select one:

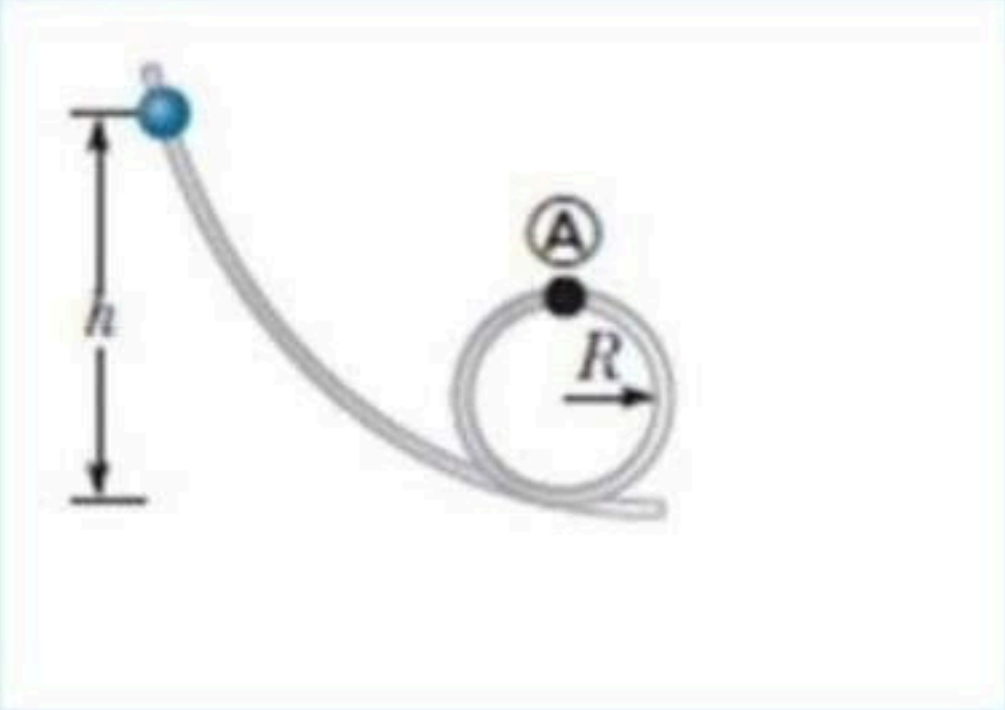
- a. 147 J
- b. -88 J
- c. 88 J
- d. -147 J
- e. 0

A tire 0.500 m in radius rotates at a constant rate of 100 rev./ min. Find the linear velocity of the tire.

Select one:

- a. **52.33 m/s**
- b. **3.14 m/s**
- c. **5.23 m/s**
- d. **2.62 m/s**
- e. **314 m/s**

A bead slides without friction around a loop-the-loop as shown in a figure below. The bead is released from rest at a height (h = 3.5R). How large is the normal force on the bead at point A if its mass is 2 grams?



Select one:

- a. 0.0392 N
- b. 0.0780N
- c. 0.9800N
- d. 0.0980 N
- e. 0.0098N

A 2.0-kg object passes through the origin at time t=0, such that its x component of its velocity is 5.0m/s and its y component of its velocity is -3.0m/s. What is the kinetic energy of the object at this time?

Select one:

a. 9J

b. 16J

c. 34J

d. 25J

e. 340J

A person weighing 0.70 kN, runs with constant speed of 7m/s, its kinetic energy (KE) is:

Select one:

a. **490 J**

b. **3430 J**

c. **17.15 J**

d. **1750 J**

e. **1.75 J**

When a car goes around a circular curve on a level road,

Select one:

a.

There is no net frictional force because the road and the car exert equal and opposite forces on each other.

b. The frictional force of the road on the car increases when the car's speed increases.

c. The frictional force of the road on the car increases when the car's speed decreases

d.

The frictional force of the road on the car increases when the car moves to the outside of the curve.

e. No frictional force is needed because the car simply follows the road.

When a car goes around a circular curve on a level road, one of the next statements is correct:

Select one:

- a. **No frictional force is needed because the car simply follows the road.**
- b. **The frictional force of the road on the car increases when the car's speed decreases**
- c. **There is no net frictional force because the road and the car exert equal and opposite forces on each other.**
- d. **The frictional force of the road on the car increases when the car moves to the outside of the curve.**
- e. **The frictional force of the road on the car increases when the car's speed increases.**

Which statement is true about the unit vectors i, j, and k?

Select one:

a. **Their directions are defined by a left-handed coordinate system.**

b. **The angle between any two is 90 degrees**

c. **Each has a length of 1 m.**

d. **i.i=j.j=k.k=0**

e. **If i is directed east and j is directed south, k points up out of the surface.**